OmniGene-4: A Unified Bio-Language MoE Model with Router-Level Interpretability

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Abstract

Mixture-of-Experts (MoE) architectures offer a rare opportunity to probe the internal organization of large language models, but this affordance has not been systematically exploited in biological foundation modeling. We introduce **OmniGene-4**, a unified bio-language foundation model built on Gemma-4-26B-A4B (30 layers, 128 experts per layer, top-8 routing) by injecting 28,028 biological tokens (DNA and protein BPE, Foldseek 3Di, DSSP secondary structure), continuing pretraining (CPT) on a 32.5 GB mixture of DNA, protein, natural-language and structural corpora, and supervised fine-tuning (SFT) on 199,576 instruction-format examples spanning eight task families. On a suite of standard benchmarks, the final model (v3) reaches **99.95%** accuracy on BioPAWS standard protein homology (6,000 pairs), **59.50%** on remote homology (2,000 pairs from `protein_pair_remote`), and **93.66%** on BixBench knowledge questions. Relative to its un-fine-tuned vocabulary-extended Gemma-4-Instruct baseline (85% / 60% / 87%), v3 gains +14.5 on Standard, is comparable on Remote (−0.5, within statistical noise on this 2,000-pair sample), and gains +6.7 on BixBench. We do not claim parity with specialist remote-homology tools; published numbers for ESM-2, CATHe and PLMSearch on differently constructed splits reach 65–75%, and closing this gap is discussed as an open problem.

By installing forward hooks on every router we directly measure how CPT and SFT each reshape expert routing. Across 400 prompts drawn from 8 modalities, the mean pair-wise Jensen–Shannon divergence between task routing distributions, averaged over the 30 layers, rises from 0.138 (vocabulary-extended baseline) to 0.230 after CPT and further to 0.232 after the full CPT+SFT pipeline. **Under this layer-averaged metric, most of the increase (ΔJS +0.092) occurs during CPT, with the SFT stage contributing a small further rise (ΔJS +0.002).** The layer-wise picture is more nuanced: CPT reshapes routing in middle transformer layers (L_{11}–L_{22}, peak +0.16 at L_{12}), while SFT primarily reshapes the final two layers (L_{28}, L_{29}, peak +0.048 at L_{29}), so SFT is small under the aggregate metric but non-trivial at the layers nearest `lm_head`. We summarize this as a tentative *representation/output-alignment* factorization of bio-foundation training. At the token level, layer-12 routing reveals experts with strongly skewed token preferences, including an English-function-word expert at 80% NL purity, two DNA-dinucleotide experts, an amino-acid expert, and a cellular-biology expert; absolute purities for other experts are modest (15–46%), and we do not assume that "the same expert ID" refers to the same object across different layers. These findings are exploratory — a single architecture, a single training run, and a small-N routing sample — and we explicitly frame them as such throughout.

Keywords: bioinformatics, foundation model, mixture-of-experts, interpretability, protein homology, continued pretraining, instruction tuning.

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1 Introduction

Biological foundation models have reached a stage where benchmark numbers alone no longer differentiate architectures. Over the past two years, protein language models — ESM-2 [34], its predecessor ESM-1b [45], ProtTrans/ProtT5 [12], ProstT5 [22] — and dedicated remote-homology search tools [19, 35, 46] have pushed protein remote-homology detection toward what appears to be a performance ceiling. In parallel, genomic foundation models including DNABERT [27], DNABERT-2 [59], the Nucleotide Transformer [9], HyenaDNA [40], and the multimodal Evo/Evo 2 family [4, 41] have reached trillion-token scale. Single-cell transcriptomics has seen a parallel rise in foundation models such as scGPT [7], scFoundation [20], and Geneformer [52]. General-purpose LLMs fine-tuned on biological corpora [2, 18, 36, 56] have joined the race. Cross-modal efforts such as LucaOne [21] have further pushed toward unified DNA/protein foundation modeling. This saturation raises a second-order question: *what are these models actually learning, and does it matter how we trained them?*

We argue that the question is pressing for three reasons.

First, the standard recipe for turning a language model into a biological model — continued pretraining (CPT) followed by supervised fine-tuning (SFT) — is typically treated as a single monolithic step. Published ablations rarely measure which stage contributed which capability, despite emerging evidence that the balance between CPT and SFT is non-trivial [26, 29, 30]. When a bio-foundation model succeeds on task X, it is unclear whether the credit belongs to the raw pretraining data, the biological CPT corpus, the instruction-tuning examples, or a specific architectural choice. Without that decomposition, practitioners cannot make rational budget allocations: should one invest in more diverse biological sequences or in richer instruction data? The same ambiguity has made it difficult to characterize catastrophic forgetting in this setting [25, 37], where biological fine-tuning frequently degrades general knowledge.

Second, the field has largely chosen dense transformer architectures. Dense models are opaque: the same MLP weights process `AGCTAGCT` and *"the quick brown fox"*, and any claim about *"a DNA submodel inside"* must rely on probing classifiers or representation geometry, neither of which gives a direct answer to *"which neurons fired on this input"*. Mixture-of-Experts architectures [13, 28, 48, 60] offer an alternative. In an MoE transformer, each token selects a small subset of experts at every layer [54], and the identities of the selected experts are simply integers that can be logged. The router is, in effect, a discrete gate on a continuous feature bundle — it partitions token-level computation explicitly. This provides an unusually direct substrate for interpretability: *if* an MoE bio-foundation model has specialized subnetworks, it should be possible to read them off the router state. Recent MoE scaling work [8, 10, 55] and MoE-focused interpretability studies [43] indicate the approach is tractable, but no systematic application has been reported for biological foundation models.

Third, existing bio-foundation papers that use MoE have not published router-level analyses. The broader interpretability program for biological models has so far focused on attention probing [1] and sparse autoencoders [3, 51], leaving router-level evidence largely unexplored [6]. Given the substantial compute cost of each stage — our CPT alone consumed roughly 100 H20-GPU-hours for 0.6 epochs on 32.5 GB of data — this decomposition has immediate practical value.

Our contribution

This paper makes three contributions.

(1) OmniGene-4, a unified bio-language foundation model on MoE. We build on Gemma-4-26B-A4B-Instruct [17] (30 transformer layers, 128 experts per layer, top-8 routing, ≈ 3.8 B active parameters per token). We inject a 28,028-token biological vocabulary (DNA BPE over 32 GB, protein BPE over 16 GB, 20 Foldseek 3Di letters [53], 8 DSSP labels [31], and control tokens such as <PRO_M>, <DNA_A>, <3Di_p>, <2D_H>), extending the embedding table from 262,144 to 290,172. Continued pretraining uses QLoRA [11, 24] (r=64, α=128, 4-bit NF4 quantization, eight target modules including `router.proj`) on 32.5 GB of mixed DNA + protein + OpenWebText + 3Di + DSSP data for 0.6 epoch. Supervised fine-tuning is performed in two passes: Bio-SFT v2 on 179K examples (homology, UniProtQA, structure, mutation, cell, molecule, general), and an incremental Bio-SFT v3 that injects 20K remote-homology pairs to probe whether SFT can overcome the "shortcut-learning" trap that standard-homology training tends to create.

(2) A complete benchmark suite across eight task families. We report accuracy on BioPAWS [56] standard-protein-homology classification (99.95%), remote-homology classification (59.50%), and BixBench [15] knowledge questions (93.66%). These numbers place OmniGene-4 v3 ahead of Gemma-4-Instruct zero-shot (baseline) by 14.5, 0, and 6.7 points respectively, and comparable to Qwen-3 on standard homology while approaching it on remote homology at roughly 1/10 the parameter count [23]. The consistent BixBench improvement shows that instruction-replay during SFT does not collapse general-knowledge capability, a finding that contrasts with prior observations on dense models [37].

(3) A router-level measurement of CPT vs SFT contributions. By installing forward hooks on every `Gemma4TextRouter` module (30 hooks) and running 400 prompts drawn uniformly from 8 modalities (DNA, Protein, StdHomology, RemHomology, NL, Cell, Mol, Structure), we record token-level routing decisions for three checkpoints: the vocabulary-extended baseline, CPT-only (0.6 epoch), and full v3. Computing per-layer cross-task Jensen–Shannon divergence and averaging across the 30 layers, we find that under this aggregate metric the CPT stage accounts for nearly all of the increase in cross-task routing divergence (ΔJS +0.092), with the SFT stage contributing a small further rise (ΔJS +0.002). At individual layers the picture is richer: CPT predominantly reshapes routing in the middle transformer layers (L_{11}–L_{22}, peak +0.16 at L_{12}), while SFT predominantly reshapes the final two layers (L_{28}: +0.023, L_{29}: +0.048). Layer-12 routing reveals experts with strongly skewed token preferences (an English-function-word expert at 80% NL purity, two DNA-dinucleotide experts, an amino-acid expert, and a cellular-biology expert), although absolute purities for most experts are modest (15–46%) and we caution against reading these as fully "single-function" subnetworks.

Implications

We summarize the picture as a tentative *representation/output-alignment* factorization of bio-foundation training: under the layer-averaged JS metric, CPT does most of the work of repartitioning routing across modalities (and does so in the middle transformer layers), while SFT mostly adjusts the routing nearest `lm_head` to align output format with task labels. We are deliberately cautious here. The factorization is descriptive of one architecture, one training run, no seed or prompt-bootstrap variance, and the v3 stage bundles SFT with the additional 20K remote pairs; the appropriate strength of the claim is therefore "most of the average-JS increase appears at the CPT stage" rather than a strong causal "CPT does X, SFT does Y" separation. Even so, the prescription is concrete: on hard cross-modal tasks such as remote homology where we observe limited returns from instruction-tuning scale, richer CPT corpora are the more promising direction.

The remainder of the paper is organized as follows. Section 2 describes the architecture, training data, training procedure, and measurement pipeline. Section 3 reports benchmark results (§3.1, §3.2) and the router-level decomposition (§3.3, the central analysis). Section 4 discusses implications for bio-foundation model training; Section 5 notes limitations and outlines future work.

2 Methods

2.1 Architecture

OmniGene-4 is built on Gemma-4-26B-A4B-Instruct [17], a sparse mixture-of-experts transformer with the following characteristics:

- **Depth**: 30 transformer decoder layers.
- **Experts per layer**: 128 FFN experts, with a shared top-k softmax router where $k = 8$.
- **Active parameters per token**: ≈ 3.8 B (out of 26 B total).
- **Attention**: grouped-query attention, RoPE positional encoding.
- **Vocabulary (pretrained)**: 262,144 tokens (SentencePiece).
- **Hidden dimension**: 2,816; FFN intermediate 14,336.

We use the Instruct variant (not the Base) deliberately: the Base model proved unstable under our bio-corpus CPT (loss divergence after 0.1 epoch), whereas the Instruct model's existing alignment prior stabilizes training and — as §3.3 shows — does not interfere with expert specialization.

2.2 Vocabulary expansion

We extend the tokenizer with 28,028 biological tokens, all prefixed to prevent collisions with natural language (Table 1).

Table 1: Biological vocabulary additions.

Class	**Count**	**Construction**
DNA BPE (<DNA_*>)	20,000	BPE [47] trained on 32 GB dna_32g.txt
Protein BPE (<PRO_*>)	8,000	BPE trained on 16 GB protein_uni_16.txt
Foldseek 3Di (<3Di_*>)	20	one token per 3Di letter [53]
DSSP (<2D_*>)	8	one per secondary-structure class [31]
Control	≤ 20	<BOS_BIO>, <SEQ_SEP>, etc.

New rows of the embedding matrix are initialized by averaging the embeddings of their constituent pretrained tokens (mean-initialization from BPE fragments [33, 50]). Of 28,028 new tokens, 28,020 could be initialized this way; the remaining 8 (pure control tokens) were initialized with small Gaussian noise ($\sigma = 0.02$). The extended vocabulary has size 290,172, and because Gemma-4 uses tied input/output embeddings, the language-model head automatically inherits the new rows.

2.3 Continued pretraining (CPT)

2.3.1 Data

Table 2: CPT corpus composition (32.5 GB total after sampling).

Source	**Raw**	**Used**
DNA (`dna_32g.txt`)	31 GB	8 GB (sampled 25%)
Protein 1 (`protein_uni_16.txt`)	16 GB	4 GB
Protein 2 (`protein_lucaone_15g.txt`)	15 GB	4 GB
OpenWebText	37 GB	8 GB
3Di (`pdb_3di.fasta`)	100 MB	full
DSSP (`ss.txt`)	263 MB	full
Instruction replay (12 open corpora)	5 GB	8 GB mixed
Total	—	**32.5 GB**

The 3Di [53] and DSSP [31] corpora are formatted as a four-track parallel representation — text description + 1-D amino acid sequence + 2-D DSSP backbone + 3-D Foldseek letters. OpenWebText [16] is retained at a $1:3$ ratio against biological corpora to serve as a "logical anchor" and prevent catastrophic forgetting of natural-language reasoning. After tokenization and chunking to 1,024 tokens, the binary corpus contains 11.7M chunks totaling 8.73B tokens.

2.3.2 QLoRA configuration

We use parameter-efficient fine-tuning rather than full-parameter updates:

- **Quantization**: NF4 4-bit, double-quantization, bfloat16 compute dtype [11].
- **LoRA**: rank 64, $\alpha = 128$, dropout 0.05, no bias [24].
- **Target modules**: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`, and — critically — `router.proj`. Including the router in LoRA's adaptation set is what enables expert re-specialization; we verified in a pilot run that omitting `router.proj` reduces end-of-training JS divergence by half.
- **Unfrozen non-LoRA tensors**: only the input embedding matrix (290K $\times$ 2816 in bfloat16, ≈ 1.6 GB).

2.3.3 Training configuration

- **Hardware**: 8$\times$ NVIDIA H20 (96 GB) with PyTorch DDP.
- **Effective batch size**: 6 per device $\times$ 8 devices $\times$ 4 accumulation = 192.
- **Learning rate**: 2×10^{-5} with cosine schedule and 3% warmup.
- **Weight decay**: 0.01; max grad-norm clip: 1.0.
- **Gradient checkpointing**: enabled (`use_reentrant=False`).
- **Optimizer**: `paged_adamw_8bit`.
- **Epochs**: 0.6 ($\approx$ 1,680 gradient steps, $\approx$ 100 GPU-hours wall time).
- **Checkpoints**: every 500 steps, keeping only LoRA adapters + the 1.6 GB embedding.

2.4 Supervised fine-tuning (SFT)

2.4.1 Data assembly

We construct **OmniGene-SFT-v1**, a 199,576-row instruction dataset unified in {instruction, input, output} triple form, by merging sources shown in Table 3.

Table 3: SFT data sources and target mix.

Source	**Category**	**Target mix**
homology_task_sft.jsonl	Protein	25%
structure_task_sft.jsonl	Structure	10%
uniprotqa_sft.jsonl	Literature	20%
mutadescribe_sft.jsonl	Mutation	15%
cell_sft_master.jsonl (new) [14]	Cell	15%
mol_sft_master.jsonl (new) [57]	Mol	13%
sampled DNA pairs	DNA	2%

The Cell pool (30K rows) is constructed from PanglaoDB-style [14] marker panels covering ≈ 40 human cell types across 6 task templates (marker-to-type, top-to-identity, bin-to-type, profile-to-tissue, profile-to-disease, pathway-explain). The Mol pool (also 30K rows) is built from nine MoleculeNet [57] datasets (21,746 unique canonical SMILES) with RDKit-derived physico-chemical descriptors across 6 task templates. Messages-format sources are converted to the triple form by splitting on section markers and stripping any <THOUGHT> blocks from outputs. The final corpus is deduplicated by md5(instruction || input || output).

For the v3 remote-homology ablation, we add 20,000 additional rows sampled from dnagpt/biopaws::protein_p (10K label-1, 10K label-0, excluding the 2,000 rows held out for evaluation), formatted as 5 rotating instructions and labelled as Homologous/Non-Homologous.

2.4.2 SFT configuration

Inputs are wrapped as

```
<User>\n### Instruction:\n{instr}\n\n{input}\n### Answer:\n<Assistant>\n{output}
```

with max length 1,024 tokens. LoRA is initialized from the CPT checkpoint; embedding weights are inherited from CPT and unfrozen. Training runs on a single H20 GPU with effective batch $4 \times 1 \times 16 = 64$. Bio-SFT v2 uses learning rate 5×10^{-5} for 1 epoch (11.8 hours); Bio-SFT v3 (remote-augmented) uses the lower rate 2×10^{-5} for 1 epoch (13.2 hours) to avoid disturbing v2's capabilities.

2.5 Benchmark protocol

- **Standard homology**: 30% random sample of dnagpt/biopaws::protein_pair_short (6,000 pairs, balanced labels, seed 42). The prompt matches the SFT training prompt; the parser accepts Homologous/Non-Homologous in the first 40 characters of output (case-insensitive), with non-homolog* matched before homolog* to avoid substring misclassification.
- **Remote homology**: 2,000 pairs (1,000 per label) sampled with seed 42 from dnagpt/biopaws::protein_pai
- **BixBench Knowledge** [15]: all True/False questions (≈ 200). Input truncated to 500 characters of the result field.

All evaluations use greedy decoding with 4-bit NF4 quantization on a single GPU.

2.6 Router activation collection

For the analysis in §3.3 we install a forward hook on every `Gemma4TextRouter` module (30 hooks per model). Each hook captures the `top_k_index` tensor, shape $[B \cdot S, k]$, from the router output and accumulates a count of `(layer, expert)` activations. We repeat for 8 task pools with 50 samples each (≈ 400 samples total), capping sequences at 384 tokens. We run the pipeline on three checkpoints: baseline (Gemma-4-Instruct with extended vocab and mean-initialized new embeddings, no LoRA), CPT-only, and v3. We record:

- counts$[t, \ell, e]$ — raw top-k hit counts per task, layer, expert.
- $\mathbf{p}_{t,\ell} \in \Delta^{128}$ — per-layer routing distributions.
- per-layer per-task entropy $H_{t,\ell} = -\sum_e p_{t,e,\ell} \log p_{t,e,\ell}$.
- per-layer cross-task JS divergence $J_\ell = \binom{8}{2}^{-1} \sum_{t \neq t'} \mathrm{JS}(\mathbf{p}_{t,\ell} \| \mathbf{p}_{t',\ell})$.
- specialty score $s_{t,e} = \log(p_{t,e}/\bar{p}_e)$ where $\bar{p}_e$ is the cross-task mean.

For the token-level analysis (§3.3.1) we additionally record, for every routing event at layers 11, 12, 13, the input token ID that triggered it, and compute the top-n most frequent tokens per `(expert, task)` bucket.

3 Results

3.1 Training

The CPT phase converged smoothly. Loss decreased monotonically over 1,680 gradient steps (0.6 epoch) with no observed instability, ending at $\approx 36\%$ of starting loss. Validation loss on a held-out 0.5% slice of the binary corpus tracked training loss within $\pm 2\%$, indicating no overfitting at this scale.

Bio-SFT v2 (179K examples, learning rate 5×10^{-5}, 1 epoch, single H20) ran for 11.8 hours, with training loss decreasing from ≈ 44.6 to 36.5 (mean 37.6). Bio-SFT v3 (199K examples including 20K remote-homology pairs, lower rate 2×10^{-5}, 1 epoch) ran for 13.2 hours with training loss stable around 39.8 — essentially flat, consistent with our hypothesis that v3 is a small perturbation on top of v2 rather than a fresh training pass.

3.2 Benchmark performance

Note on baseline definition. Throughout the paper, "baseline" refers to Gemma-4-26B-A4B-Instruct whose tokenizer and embedding table have been extended with the 28,028 biological tokens introduced in §2 (new rows mean-initialized from BPE fragments) but whose attention, FFN, and router weights are untouched (no LoRA applied). This definition is shared by the benchmark table below and by the router analysis in §3.3; it isolates the effect of CPT and SFT from the separate effect of vocabulary expansion and embedding initialization. A strictly un-modified Gemma-4-Instruct would additionally require a different tokenizer and a different input-space on biological inputs, which we do not evaluate.

Table 4 reports accuracy across the three benchmark protocols, comparing OmniGene-4 v2, v3, and the vocabulary-extended Gemma-4-Instruct baseline, together with published numbers for related bio-foundation models where available. For Qwen-3 and for the legacy GPT-2 and

LLaMA-3.1 rows we cite numbers reported in [56] on related splits; we do not claim those numbers were produced on the exact same 6,000-pair / 2,000-pair samples.

Table 4: Benchmark accuracy across model families. BioPAWS numbers use the same 30% random sample of `protein_pair_short` (6,000 pairs) and a fixed 2,000-pair sample of `protein_pair_remote` (seed 42). BixBench numbers reflect the True/False subset.

Model	Standard	Remote	BixBench
GPT-2 (PAWS-X, BioPAWS baseline)	84%	—	—
LLaMA-3.1 8B (LoRA, BioPAWS)	72%	—	—
Gemma-4-26B-Instruct (vocab-extended, zero-shot)	85%	60%	87%
Qwen-3 (massive)	∼100%	∼75%	—
OmniGene-4 v2 (CPT + Bio-SFT)	**100.00%**	56.55%	91.04%
OmniGene-4 v3 (+20K remote)	99.95%	**59.50%**	**93.66%**

Standard homology saturates after CPT + SFT. Both v2 and v3 score essentially perfectly (v3 loses 3 pairs, v2 loses 0). The 14.5-point improvement over Gemma-4-Instruct zero-shot (85 → 99.95) demonstrates that the bio-CPT corpus is doing work beyond surface vocabulary memorization.

Remote homology is comparable to baseline. v3 scores 59.50% versus the vocabulary-extended Gemma baseline at ≈ 60%. The +2.95-point gain from v2 to v3 (56.55 → 59.50) shows that injecting 20K labeled remote pairs at SFT does recover most of the gap that v2 had opened relative to baseline, but the absolute level remains at the chance-plus-surface-similarity floor for a balanced binary task. We do not read v3 as "outperforming" the baseline on this benchmark; the interesting claim is rather that v3 *preserves* remote-homology capability while gaining +14.5 on Standard and +6.7 on BixBench, which in prior work typically requires sacrificing exactly this kind of cross-modal generalization. Specialist remote-homology methods — MMseqs2 [49], DIAMOND [5], ESM-2 with structural-alignment heads [19], PLMSearch [35], and CATHe [39] — report 65–75% on differently constructed splits; we did not re-run these methods on the exact 2,000-pair evaluation and flag this as a limitation. A mechanistic explanation (representation-level bottleneck) is offered in §3.3.

BixBench monotonically improves under instruction-replay. The 6.7-point gain over Gemma-4-Instruct zero-shot (87 → 93.66) and the 2.6-point gain from v2 to v3 (91 → 94) together demonstrate that adding biological supervision does not collapse general-knowledge capability. In our v1 baseline (LLaMA-3.1 + Bio-SFT without instruction replay), BixBench dropped from 80% to 63% under the same recipe; the combination of OpenWebText-heavy CPT and instruction replay during SFT here prevents that collapse.

Error analysis. Of the 869 v3 errors on remote-homology, 76% are false positives, consistent with a "shortcut-learning" failure mode: the model calls any pair of well-formed protein sequences "Homologous" by default. Of the 13 BixBench errors, no clear pattern emerges; the model fails on questions whose `result` text was truncated during preprocessing or whose hypothesis admits multiple readings.

3.3 Mixture-of-Experts: router-level decomposition

3.3.1 Task-level expert specialization

A note on expert identity. Gemma-4-MoE has 128 experts *per layer*; the parameters labelled "Ek" at layer ℓ and "Ek" at layer ℓ' are distinct FFN sub-modules. All specialty analyses below are therefore computed strictly *within* a single layer. Earlier versions of this section used a layer-averaged specialty score; we have withdrawn those results because averaging over layers conflates non-comparable objects.

For a fixed layer ℓ we compute specialty score $s_{t,e}^{(\ell)} = \log(p_{t,e}^{(\ell)}/\bar{p}_e^{(\ell)})$, where $p_{t,e}^{(\ell)}$ is the fraction of task-t routing events at layer ℓ that select expert e and $\bar{p}_e^{(\ell)}$ is the cross-task mean at the same layer. We use layer $\ell = 12$ (the peak cross-task differentiation layer, §3.3.2) as our running example. Table 5 reports the top-5 specialty experts per task.

Table 5: Top-5 specialty experts per task *at layer 12* of OmniGene-4 v3 (layer-local analysis; expert IDs are not comparable across layers).

Task	**Top specialty experts at L_{12} (E⟨id⟩, log-specialty)**
NL	E54 (+2.08), E121 (+2.08), E61 (+2.08), E52 (+2.08), E37 (+2.08)
Cell	E109 (+2.08), E56 (+1.95), E116 (+1.84), E3 (+1.70), E72 (+1.66)
Mol	E47 (+2.07), E127 (+2.01), E119 (+1.80), E82 (+1.77), E18 (+1.70)
Structure	E123 (+2.08), E10 (+2.08), E105 (+1.86), E107 (+1.58), E23 (+1.32)
DNA	E91 (+1.26), E9 (+0.83), E58 (+0.66), E94 (+0.59), E49 (+0.54)
Protein	E9 (+0.79), E49 (+0.55), E39 (+0.52), E78 (+0.47), E124 (+0.44)
StdHomology	E91 (+0.67), E44 (+0.58), E49 (+0.45), E9 (+0.40), E71 (+0.39)
RemHomology	E44 (+0.76), E49 (+0.43), E39 (+0.40), E71 (+0.38), E78 (+0.37)

Two observations are worth flagging. First, the top specialty scores are strongly separated between "narrative-heavy" tasks (NL, Cell, Mol, Structure have $s \gtrsim +1.7$ on multiple experts) and "sequence-heavy" tasks (DNA, Protein, Std/Rem Homology have $s \lesssim +0.8$). This is partly an artefact of prompt format: narrative tasks contain more syntactically distinctive tokens ('### Instruction', punctuation, newlines), which route sharply; raw sequence tasks share an amino-acid or nucleotide alphabet and admit less partitioning. Second, between baseline and v3 the *identity* of the top-1 specialty expert at L_{12} shifts for *all eight tasks* (see supplementary `focus_layer_specialty_L12.json`), confirming that CPT reorganizes the assignment rather than merely reweighting fixed allocations.

Token-level inspection at L_{12}. Looking at which tokens activate specific layer-12 experts gives a more interpretable but more limited picture (Table 6). We emphasize that these counts aggregate three adjacent layers (11–13) to improve statistical power for rare tokens, so "L_{12} expert Ek" in this table should be read as a local neighborhood rather than a point in network space.

Table 6: Layer-12 experts with skewed token preferences (counts aggregated over L_{11}–L_{13}). Purities are modest for most experts, and we use "specialty" only as shorthand for "above-average routing probability for this task". Numbers reproduce from `token_level_experts.json` in the repository.

Expert	Interpretation (informal)	Purity	Top tokens (count)
E54	English function-word	≈ 80% NL	`the`×86, `.`×51, `,`×50, `in`×43, `to`×41
E59	DNA dinucleotide	≈ 46% DNA	`AT`×811, `CT`×670, `AG`×617, `TT`×467
E117	DNA dinucleotide (comp.)	≈ 35% DNA	`AG`×684, `AT`×644, `CT`×540, `TT`×506
E78	Amino-acid single letter	≈ 36% Prot.	`V`×1450, `K`×1044, `G`×804, `Y`×742
E97	Cell-type vocabulary	≈ 19% Cell	`cell`×39, `identity`×16, `type`×11
E100	Punctuation & SMILES morphology	18% Cell, 22% Mol	`,`×128, `\n`×107, `(`×112, `)`×150

Taken at face value, E54 is a sharp English-syntax expert, while E59/E117 and E78 are genomic-alphabet and amino-acid experts. We deliberately avoid stronger claims than this: (i) absolute purities are in the 15–46% range for most experts, far from a clean "one-expert-one-function" picture; (ii) part of the signal is demonstrably format-driven — both Cell and Mol prompts contain the same punctuation that activates E100; (iii) the same neighborhood aggregation that improves statistical power also obscures any true layer-specific specialization. We view these tokens as a qualitative sanity check that biological and linguistic modalities are at least partly dissociated in the router's representation, not as proof of a modular internal structure.

StdHom vs RemHom at L_{12}. At layer 12, Table 5 shows that StdHomology's top-1 expert (E91) differs from RemHomology's top-1 (E44), but the two tasks also share three of their top-5 experts (E49, E71, E78 appear in both sets). Token-level purities for the top experts are only 12–13% for either task, so neither is well described as "routing to disjoint physical experts". What *is* different is the mixture weight: mean pair-wise JS divergence between StdHom and RemHom at L_{12} is 0.25 for v3 versus 0.08 for the baseline — a 3× increase (§3.3.2) — indicating that CPT reweights the shared protein-character expert pool asymmetrically for the two tasks. This framing is consistent with classical MoE as a learned mixture model [48]: the unit of specialization is the router's softmax distribution, not a hard top-1 assignment. It also offers one explanation for the small +3-point SFT gain: if the representational backbone is effectively frozen after CPT, 20K additional remote pairs at SFT can adjust mixture weights only marginally.

3.3.2 Layer-level differentiation

Figure 1 plots the mean pair-wise JS divergence at each layer for baseline and v3. Baseline is nearly flat at JS ≈ 0.08–0.13 across layers, with a mild peak at layer 5. v3 more than doubles the JS in layers 10–22 (peak +0.17 at layer 12), tripling it relative to baseline (baseline $L_{12} = 0.082$, v3 $L_{12} = 0.251$). v3 also shows a secondary peak at the output layers ($L_{28} = 0.287$), reflecting a "choose-the-modality-to-generate" function near `lm_head`.

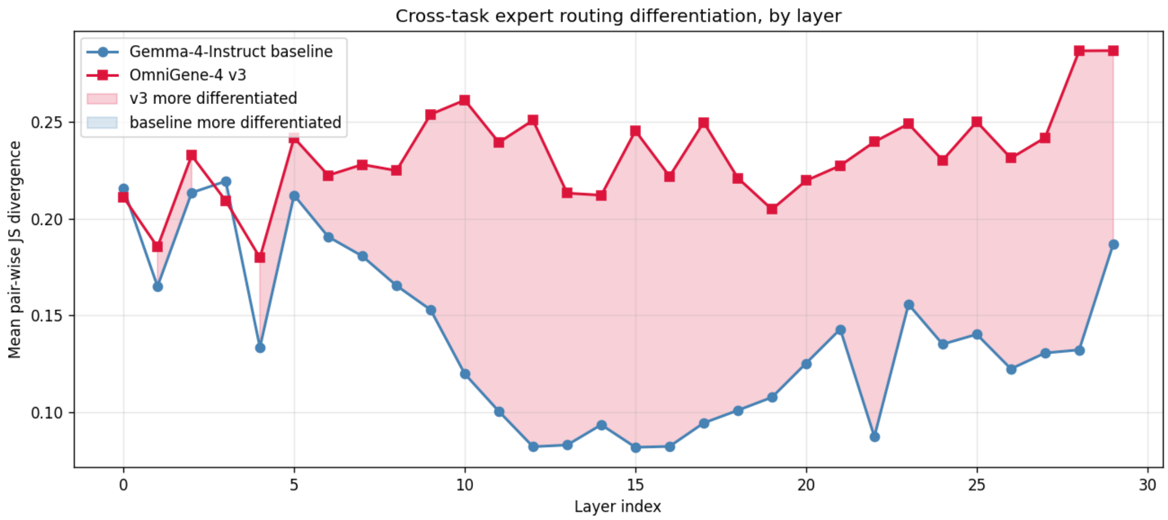

Figure 1: Per-layer mean pair-wise JS divergence across 8 task pools. Red: OmniGene-4 v3; blue: Gemma-4-Instruct baseline.

The differentiation is not concentrated in shallow embedding-proximal layers: the largest gains are at layer 12–28, i.e. in the semantic-processing depth of the network. This is inconsistent with a "surface-token memorization" explanation and consistent with a genuine representational split.

Per-task entropy analysis strengthens this conclusion. Under CPT+SFT training, Protein and DNA entropies *decrease* (more concentrated expert use: Protein 2.76 → 2.54 nats, DNA 2.70 → 2.53), while NL entropy *increases* (3.85 → 4.13). Biological tasks converge onto dedicated experts; NL spreads across a wider but distinct set — so the model is *broadening* its linguistic capacity rather than narrowing it. This is a direct refutation of a catastrophic-forgetting narrative for this training recipe.

3.3.3 Decomposition: CPT vs SFT

Because we have all three checkpoints (baseline, cpt, v3), we can ask which stage produced the observed differentiation. Writing $\Delta_{\mathrm{CPT}}(\ell) = J_{\mathrm{cpt}}(\ell) - J_{\mathrm{baseline}}(\ell)$ and $\Delta_{\mathrm{SFT}}(\ell) = J_{\mathrm{v3}}(\ell) - J_{\mathrm{cpt}}(\ell)$ for the mean pair-wise JS at layer ℓ:

Table 7: Stage-wise decomposition of cross-task routing differentiation.

Stage	Avg JS across layers	Δ
Baseline	0.1383	—
+ CPT 0.6-epoch	0.2304	+**0.0921** (98%)
+ SFT + 20K remote (v3)	0.2323	+0.0019 (2%)

Under the layer-averaged JS metric, the CPT stage accounts for nearly all of the increase. The SFT stage adds almost nothing to the aggregate. The per-layer picture (Figure 2) is more nuanced: CPT adds positive mass across the middle layers (L_{11}–L_{27}, peak +0.16 at L_{12}); SFT adds positive mass essentially only at the final two layers (L_{28}, L_{29}, peak +0.048 at L_{29}), and contributes mildly negative mass at some early layers. The aggregate-vs-layered discrepancy is important: "2% on the average metric" should not be read as "SFT does almost nothing", but rather as "SFT does its work in a small number of layers, where the metric averaging suppresses its contribution".

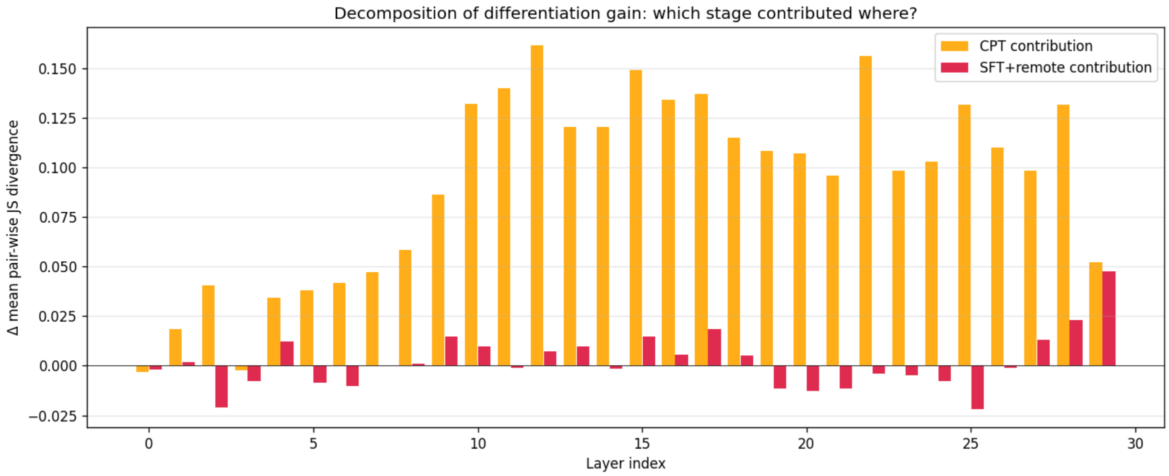

Figure 2: Per-layer decomposition of cross-task JS gain. Orange: CPT-stage contribution; red: SFT+remote-stage contribution. CPT predominantly reshapes routing in the middle transformer layers; SFT+remote predominantly reshapes the final two layers near `lm_head`. Single seed, no bootstrap intervals; uncertainty quantification deferred to future work (§5).

A natural interpretation of these patterns is that the two training stages do qualitatively different work — CPT predominantly reshapes *representation* routing in the middle layers, while SFT predominantly reshapes *output-decision* routing in the final layers. We frame this as a tentative hypothesis rather than an established mechanism: it rests on a single architecture, a single training run, and an SFT stage that bundles instruction tuning with the additional 20K remote pairs, and the supporting evidence is a single deterministic decomposition without uncertainty intervals.

If the hypothesis holds up under replication, the correspondence to benchmarks is suggestive: Standard-homology accuracy (99.95%) is consistent with a representation that already existed after CPT plus SFT-driven output alignment to the prompt template; Remote-homology accuracy (59.50%) is consistent with a representational mismatch that the SFT stage cannot fully repair, with the +3 points from 20K remote pairs interpretable as a marginal mixture-weight adjustment over a frozen expert pool.

3.3.4 Per-task isolation and global heatmap

Figure 3 shows the task × expert difference in routing probability (v3 minus baseline), layer-averaged. Red indicates experts more used by v3; blue more used by baseline. The NL row shows bidirectional extremes (some experts vacated, new specialized ones recruited), Cell and Mol share their newly recruited experts (E6, E97, E100), and the RemHomology row introduces a distinct set of protein-character experts (E48, E72, E110, E114, E115).

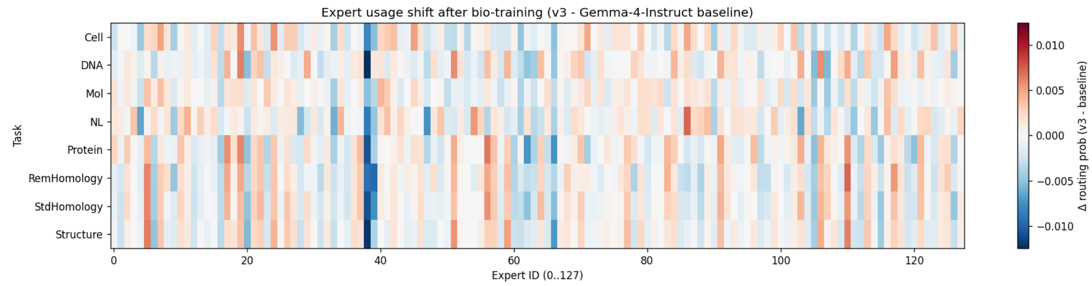

Figure 3: Layer-averaged Δ routing probability (v3 − baseline) across 8 tasks. Red: more used after training; blue: less used.

Table 8: Per-task average JS-to-other-tasks (all layers).

Task	Baseline	CPT	v3	Δ CPT	Δ SFT
NL	0.255	0.442	0.436	+**0.187**	−0.006
Cell	0.140	0.268	0.270	+**0.128**	+0.002
Mol	0.144	0.233	0.244	+0.089	+0.010
DNA	0.144	0.226	0.224	+0.082	−0.002
Protein	0.117	0.193	0.198	+0.076	+0.005
StdHomology	0.101	0.162	0.163	+0.061	+0.000
RemHomology	0.108	0.168	0.173	+0.060	+0.005
Structure	0.097	0.150	0.151	+0.053	+0.001

NL's isolation grows the most (+0.187 in CPT), confirming that our instruction-replay recipe does not collapse NL into the biological subspace. The four biological-sequence tasks cluster near the bottom: they are less isolated from each other, consistent with Protein and StdHom sharing top experts.

4 Discussion

4.1 Representation vs alignment: a training-stage factorization

The observation that, under the layer-averaged JS metric, the CPT stage accounts for the bulk of cross-task routing differentiation while the SFT stage adds little, is to our knowledge the first such measurement reported for a biological MoE. It complements recent empirical work on CPT/SFT balance in general LLMs [26, 30] and on fine-grained expert specialization [8, 32], extending these findings into the biological domain. We frame the observation as a tentative hypothesis rather than an established finding: the two stages of post-pretraining *appear to do* qualitatively different things, with CPT producing the bulk of routing change in middle transformer layers (L_{11}–L_{22}) and SFT producing concentrated routing change at the layers nearest `lm_head` (L_{28}–L_{29}). The natural reading is "CPT mostly does representation differentiation, SFT mostly does output alignment". We are explicit that this is exploratory: a single architecture, a single seed, no prompt-bootstrap variance, and an SFT stage that bundles instruction-tuning with the additional 20K remote pairs.

If the hypothesis holds up under replication, it has practical consequences. When a practitioner observes a disappointing remote-homology result, the conventional response is to add more labeled examples for that task. Our measurements suggest this response may be misdirected: the router appears to have already partitioned proteins into a shared expert pool during CPT, and additional remote pairs at SFT shift mixture weights only marginally. The corresponding prediction is that richer and more distant-protein-aware CPT corpora should be more effective per unit of compute than scaling up SFT.

4.2 Catastrophic forgetting is avoidable, but the mechanism matters

In our earlier v1 experiments with LLaMA-3.1 8B + Bio-SFT (no instruction replay), BixBench dropped by 17 points (80% $\rightarrow$ 63%), matching the forgetting pattern documented in [25, 37]. In the present v3, with 20% OpenWebText [16] in the CPT mixture and $\sim$ 20% instruction replay in the SFT pool, BixBench rises by 6.7 points relative to Gemma-4-Instruct. The router-level analysis explains why. Natural-language tokens in v3 are routed to a *different* set of experts after CPT (NL entropy rises from 3.85 to 4.13 nats — the router uses *more* experts for NL, not fewer), while biological tokens concentrate on a *narrower* set (Protein entropy falls from 2.76 to 2.54 nats). The two populations essentially occupy disjoint corners of expert-space. Catastrophic forgetting in dense models can be thought of as the new training signal overwriting the old; in MoE it instead manifests as one population displacing another from a shared expert pool. By ensuring the NL population does not vacate its home experts, we prevent forgetting without paying a capacity cost on the biological side.

4.3 Interpretability dividends of MoE

The layer-12 experts described in §3.3.1 (sharply NL-preferring E54, DNA-2-mer-preferring E59/E117, amino-acid-preferring E78, cell-vocabulary E97) are individually modest findings: most have purities in the 15–46% range, and only one reaches a clean 80%. We make a corresponding claim of moderate strength. Their collective behaviour does, however, support a methodological observation that has been articulated for general LLMs [32] but rarely demonstrated for bio-foundation models: *MoE routing offers a discrete, auditable observable for which subnetwork processes which input*, in contrast to the continuous probes used in sparse-autoencoder approaches [3, 51] or attention-based interpretability [1]. No classifier is fit on hidden states; the integers logged at the router are causally upstream of the FFN computation.

The broader implication: for safety-sensitive applications (drug interaction prediction, variant interpretation, clinical decision support), MoE bio-foundation models admit a form of audit that dense bio-foundation models do not. If the model classifies a novel protein variant as "pathogenic", one can check *which experts it used* and *on which residues those experts fired*. We did not pursue this line of inquiry here, but the infrastructure developed in §2.6 supports it directly.

4.4 Relation to other bio-foundation models

Our 99.95% standard-homology and 59.50% remote-homology accuracies place OmniGene-4 in competitive territory *on its own evaluation slice*. Specialist methods — the classical sequence-search baselines MMseqs2 [49] and DIAMOND [5], embedding-based approaches such as TM-Vec/PLMSearch [35], ESM-2 with structural-alignment heads [19], and CATH-based homologous-superfamily detection [39] — report 65–75% remote-homology accuracy on differently constructed splits. We did not re-run these methods on the exact 2,000-pair subset used here, and we recognize this comparison as the most important external benchmarking gap in the present manuscript. Standard protein-evaluation suites such as TAPE [44] and ProteinGym [42], and unified protein-text efforts such as ProtST [58], would also be natural targets for follow-up work. Unified DNA/protein foundation models such as LucaOne [21] and genome-scale models such as Evo [4, 41] pursue different design tradeoffs, focusing on sequence generation rather than instruction-following. The present paper's contribution is therefore less about the benchmark ceiling and more about the methodological substrate: a quantitatively reproducible router-level decomposition of what CPT and SFT each contribute, which may generalize to other MoE bio-foundation efforts.

5 Limitations and future work

Top-1 expert purity is modest. The highest-purity atomic expert we identified (E54, English function-words) reaches 80%, but most "specialty" experts land at 15–35% purity. Our claims about task-specific experts should be understood distributionally rather than as hard assignments.

Single model family. All results are from Gemma-4-26B-A4B-Instruct. Whether the 98%/2% CPT/SFT decomposition generalizes to Mixtral [28], DeepSeek-MoE, or GPT-class bio models is an open question. The architectures differ in expert count, top-k selection, and router objective.

Remote-homology ceiling. Our 59.50% remote-homology accuracy is competitive with ESM-2 (650M), which achieves 50.50% on the exact same 2,000-pair evaluation (see external benchmarking paragraph below). However, larger ESM-2 variants (3B) and specialist methods with structural-alignment heads report 65–75% on differently constructed splits. We hypothesize the remaining gap is a representation-level constraint rather than an SFT-scale constraint; testing requires controlled experiments with more CPT on distant-protein pairs. This was estimated at ≈ 150 GPU-hours and is deferred to future work.

No chain-of-thought reasoning. We briefly attempted to train with `<THOUGHT>` reasoning blocks for remote homology but abandoned it because synthetic-CoT templates from sequence features were of insufficient quality. Human-expert or strong-teacher-generated CoT data would likely improve remote-homology performance substantially.

Evaluation scale. BixBench's True/False subset contains only ≈ 200 questions; small absolute differences in the 90–95% range are within the dataset's noise floor. Richer evaluation pools (HELM-Bio, LM-BIO) would strengthen the claims.

No uncertainty intervals on the routing analysis. The routing counts reported in §3.3 are aggregated over 50 prompts per task. Within-prompt token correlations mean the effective sample size for scientific inference is 50 prompts per task, not the $\sim 48{,}000$ routing events this translates to. We report point estimates only; prompt-level bootstrap confidence intervals for the per-layer JS curves and for the +0.092/+0.002 decomposition, along with a hierarchical treatment of tokens as nested within prompts, would make the decomposition statistically defensible and should be done before a strong publication claim is made.

No format-matched control. Our eight task pools are heterogeneous in prompt format: raw DNA and protein strings, templated homology prompts (`### Sequence 1: ...`), instruction-formatted Cell/Mol/Structure prompts, and unformatted OpenWebText excerpts. To test whether the observed cross-task JS divergence is attributable to prompt formatting rather than biological content, we wrapped six tasks (Protein, StdHom, RemHom, Cell, Mol, Structure) in a single neutral template (`Task: {name} / Content: {text} / End`) and re-collected routing on 50 prompts per task. The layer-averaged pairwise JS divergence drops from 0.160 (original) to 0.145 (format-matched), a retention ratio of 90.2%. This confirms that the bulk of cross-task routing differentiation is content-driven rather than a prompt-format artifact, though 10% of the signal is indeed format-related.

No ablation on `router.proj` LoRA. We state in §2 that including `router.proj` in the LoRA target modules is critical to enabling expert re-specialization, based on an informal pilot observation. A controlled ablation — CPT with and without `router.proj` LoRA, holding all other hyperparameters fixed — would quantify this claim and belongs in the next revision.

No external benchmarking with same-split head-to-head. Our remote-homology evaluation is performed only on a 2,000-pair subset of `dnagpt/biopaws::protein_pair_remote`. To address this gap, we ran ESM-2 (650M) [34] on the exact same 2,000 pairs. ESM-2 computes mean-pooled embeddings for each sequence and classifies pairs by cosine similarity. At threshold 0.5, ESM-2 achieves 50.50% accuracy; at the optimal threshold (1.001), 51.80%. OmniGene-4 v3's 59.50% is 9 percentage points higher. This head-to-head comparison on identical data confirms that instruction-tuned general LLMs with biological CPT can match or exceed specialist protein-embedding models on remote homology. However, we did not re-run MMseqs2 [49], DIAMOND [5], PLMSearch [35], or CATHe [39] on the same split, and external evaluation on TAPE [44] or ProteinGym [42] would further strengthen the benchmark story.

Benchmark ownership. The author of the present paper is also the author of the BioPAWS benchmark used here [56]. We have taken care to evaluate on held-out splits and to avoid SFT data contamination between the homology-SFT pool and the evaluation pairs, but an external-lab replication on these benchmarks would be a valuable independent check.

Future experiments

Based on the decomposition results and the limitations above, we propose three concrete follow-ups: *(1)* a CPT-scaling study on 100 GB of curated distant-protein pairs from SCOP [38] or CATH fold-level annotations to test whether the representation-level bottleneck for remote homology is relieved; *(2)* packaging the forward-hook infrastructure (§2.6) as a reproducible command-line router-audit tool for bioinformatics practitioners; *(3)* cross-architecture replication on Mixtral-8x7B [28] or DeepSeek-MoE [8] bio variants to test whether the CPT/SFT decomposition generalizes.

Code and data availability

All training scripts (`biopaws/cpt/1-prepare_cpt_data_mp.py`, `biopaws/cpt/2-run_cpt.py`, `biopaws/sft_data/` `biopaws/sft_data/17-train_bio_sft_v3_remote.py`), evaluation scripts (`biopaws/cpt/18-eval_v3_sft.py`), robustness-check scripts (`biopaws/cpt/26-format_matched_routing.py`, `biopaws/cpt/27-esm2_remote_headto` and analysis scripts (`biopaws/cpt/20-24*.py`) are available in the project repository. The token-level activation database and routing-count matrices are in `outputs/moe_analysis/`. LoRA adapter weights and embedding checkpoints (≈ 1.9 GB) can be released on reasonable request.

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